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Training Meta Neural Networks Using Data Generated From Agent Based Models

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Abstract

Agent based models (ABMs) offer a flexible and interpretable alternative to mathematical models such as differential equations for modeling complex systems with interacting parts. These features make them an attractive option for simulating data for meta learners. This study aimed to assess the viability of this technique, comparing performance to standard neural networks, and traditional mathematical models. We found that agent based meta learners (ABMLs) significantly outperformed standard neural networks, but were not as accurate as mathematical models. These findings demonstrate the potential of ABMLs as a versatile modeling framework for limited data environments such as epidemiology, finance, businesses analytics, economics, policy making, and other social sciences.

Keywords: Metal Learning, MAML, Agent Based Models

1 Introduction

The gradient based training of neural networks such as multilayer perceptron (Rosenblatt, 1958), convolutional neural networks (LeCun et al., 1998), recurrent neural networks (Rumelhart et al., 1986), transformers (Vaswani et al., 2017), residual neural networks (He et al., 2016), and autoencoders (Hinton and Salakhutdinov, 2006) has been established as the state of the art technique for learning complex tasks. These architectures have found applications such as image recognition (He et al., 2016), natural language processing (Vaswani et al., 2017), time series analysis and financial forecasting (Shen et al., 2012), credit scoring, and video analysis (Tran et al., 2015).

Common to many deep learning techniques, particularly those based on neural networks, is the large number of parameters, and the corresponding large amount of data required to train the networks. This is a significant limitation that is being addressed in the field of few shot learning. The goal of few short learning is to replicate the ability that humans have to learn something from only a small number (as little as 1) of examples (Vinyals et al., 2016). While there has been much success in few shot learning, typically a large collection or related data sets are needed. Few shot learning, or at least limited data learning, is a common task with examples including disease forecasting, price elasticity models, macroeconomic policy models, and marketing models. These tasks are often limited to a few years of weekly or quarterly data, and often do not have a large variate of related data sets to apply few shot learning techniques to. As such these fields have seen limited success in using deep learning or few shot learning techniques, primarily resorting to traditional mathematical models such as differential equations, difference equations, and linear regression models (Petters, 2010; Davis, 2011; Joshi, 2003; Kopp et al., 2009). These mathematical models are often limited in their complexity to develop and difficulty to explain.

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In some domains, where the system is well understood, this data limitation can be overcome by simulating data and using this simulated data to train few shot models. An example of this is in supernova classification. Modern astronomical surveys such as Collaboration (2024) are set to observe $250,000$ supernova per year. The supernova need to be classified according to their type. Because the physics of supernova are relatively well understood, classifiers have been developed on simulated data, and deployed on real world data, such as that developed by Möller and de Boissière (2020).

However, in may other domains, this method become impractical. An example is in disease modeling where to simulate data a mathematical model such as the SIR model is needed, and typically this model can simply be fit to data itself (Kermack and McKendrick, 1927).

One particularly effective way of simulating data from systems such as in disease modeling, where the model itself cannot simply be fit to data, is an "agent based model" (ABM). ABMs are applied to systems where there are individual units (agents e.g. people, companies, animals, countries, particles) where the individual response of each agent is understood, but the overall dynamics of a system of agents (the population) is complex. ABMs have a few attractive features compared to mathematical models like differential equations (henceforth mathematical model will be used as a general term for ODEs, PDEs, OLS, etc. and abbreviated MM). Firstly, they are often relatively simple to define, and can be easily explained in a non-technical way to non-mathematicians. Secondly, they can model complex interactions far more effectively than mathematical models in certain circumstances. Thirdly, the rules for the agents can be tested in small scale, comparatively inexpensive, controlled experiments (Macal and North, 2006). One application of ABMs is for interacting particles where it is easy to describe how two particles should interact with each other. However, a collection of particles, a fluid, has far more complex dynamics. ABMs are thus often used for fluid simulations (Chorin, 1968). Another common application of ABMs is in the social sciences, where it is easy to describe how an individual may respond to a policy, but understanding how an entire society will respond is complex (Macal and North, 2006).

A particular domain of few shot learning is "meta learning" or "learning to learn" (Finn et al., 2017; Arnold et al., 2020). While there are many meta learning techniques, the general procedure involves training a "meta model" on a series of related tasks. The job of the meta model is to ingest a small set of training data and produce a "base learner" model that can adapt easily to the specific task. For example, the learning task may be two class image classification. The meta learner is trained on a large number of tasks, where each task consists of a small (potentially only 1 image per class) training set (known as the support set), and a similar sized testing set (known as the query set), these tasks could be, for example, classifying a cat vs a dog, a car vs a plane, a lion vs a tree etc. Once trained, the meta model could receive a small data set of images that it has never seen before, such as a book vs a lamp, and produce a base learner that can be adapted on this small data set to classify books and lamps (Arnold et al., 2020).

More precisely, meta learning consists of a distribution of tasks $P(\mathcal{T})$, each task a specific problem. Each task as a support set $S = \{(x_1, y_1), (x_2, y_2), (x_3, y_3), ...\}$ and a query set $Q = \{(x_1, y_1), (x_2, y_2), (x_3, y_3), ...\}$. There is some base learner f_θ that is adapted to each task $\mathcal{T}_i$ to get $\theta_i = \text{Adapt}(f_\theta, S_i)$. The Adapt step could be any learning process, but is often a few steps of gradient decent. The meta learner is tasked with learning the best

parameters ϕ for the Adapt process, that is to find

$$\min_{\phi} \mathbb{E}_{\mathcal{T}_i \in p(\mathcal{T})}[\mathcal{L}(f_{\theta_i}, Q_i)]$$

(Finn et al., 2017)

A number of different meta learning algorithms have been proposed. One of the most popular is the "Model-Agnostic Meta-Learner" (MAML). Its popularity is in part due to the fact that the algorithm works on any model that can be trained with gradient decent (Finn et al., 2017). Briefly, MAML works by finding the optimal initial parameters θ_0 that can be quickly adapted to a specific task.

The aim of this paper is to viability of using an ABM to simulate a collection of tasks to train a meta model on. It is expected that the agent based meta lerner (ABML) should outperform a standard neural network, and may be able to reach performance similar to a tailor made mathematical model under certain circumstances.

2 Method

To test the concept, a sample problem has been proposed. This problem has been chosen due to its simplicity, and the fact that there is a closed form mathematical model that describes the system. The performance for the ABML will be compared to both the mathematical model and a standard NN with the same architecture as base learner.

2.1 The Sample Problem

The sample problem is a simple marketing problem. There is some global population, each day some subsample of this population is taken and shown an ad. On being shown the ad, a person either converts or doesn't with some constant probability. Once a person has converted, they will no longer convert again. Once a person has converted they will continue to buy the product *ad infinitum*, as such one sale is recorded each day for them. While this simple model is by no means an accurate description of how marketing works, for its simplicity it captures the idea of "market saturation", a key concept in marketing. It is also easy to imagine how this could be extended to include concepts such as customer churn, word of mouth, media buy in, ad targeting, etc.

Given that once converted the user continues to buy the product, the sales per day is simply going to be the total number of conversions.

2.2 The ABM

Parameters P the population, S the subsample size, and c the conversion rate are chosen such that $S < P$ and $c \in (0, 1]$

The population is initialized as some vector $p = [0, 0, 0, 0, ...]^T$ of size P. Each day a subsample of S integers I between 0 and P is generated without replacement. We also generate some vector of P random numbers C from an independent uniform distribution between 0 and 1. We set all elements of p in $\{p_i \in p \mid i \in I \text{ and } C_i < c\}$ to 1. That is we randomly convert everyone in our subsample with probability c. The total sales are recorded each day as $T = \sum_i p_i$.

10,000 samples were simulated from the ABM. Each sample had 15 time steps, an integer population P sampled from a uniform distribution over $[0, 1000)$, a subsample size $S = \alpha P$ for $\alpha \in (0, 1]$ uniform, and a conversion rate $c \in (0, 1]$ uniform. The first 10 time steps were used for training (or support), and the last 5 for testing (or query). The order was maintained to ensure causality.

2.3 ABML

The base learner was a neural network with one input node (time), one hidden layer with 5 nodes, and one output node (sales for that time). The base learners "adapt" process was gradient decent with a learning rate of 0.01 and 15 decent steps.

The meta learning algorithm chosen was MAML. The meta learning rate was 0.001. The meta model was trained on all but 100 of the 10,000 samples. The renaming 100 were used as the "test" samples. Each task had a support set of size 10, and a query set of size 5 as outlined above.

2.4 Mathematical Model

The system is described by

$$T(t+1) = T(t) + p \cdot \frac{S}{P}\left(P - T(t)\right) \tag{1}$$

With initial condition $T(0) = 0$. Converting to continuous time we have

$$\frac{dT}{dt} = p \cdot \frac{S}{P}(P - T(t)) \tag{2}$$

Which has solution

$$T(t) = P\left(1 - e^{-p \cdot \frac{s}{P} t}\right) \tag{3}$$

Which we will write as

$$T(t) = P\left(1 - e^{-\alpha t}\right) \tag{4}$$

This was fit to each of the 100 held out tasks using OLS.

2.5 Standard NN

The standard NN chosen was identical to the base learner for the ABML. The model was trained for 500 epochs.

3 Results

For sample i of the 100 test samples, y_i was the actual total sales in the 5 day holdout period and $\hat{y}_i$ was the predicted total sales in the 5 day holdout period. The following statistics were calculated across the 100 test samples.

	NN	ABML	MM
MPE	22±4%	-12±5%	-1.7±1.0%
MAPE	27±3%	22±4%	3.4±0.8%

Table 1: Performance metrics for the standard neural network (NN), the agent based meta learner (ABML), and the mathematical model (MM).

$$MPE = \frac{1}{n} \sum \frac{y_i - \hat{y}_i}{y_i} \tag{5}$$

$$MAPE = \frac{1}{n} \sum \frac{|y_i - \hat{y}_i|}{y_i} \tag{6}$$

The standard error of these estimates was calculated by bootstrapping. The results are shown in Tab. 1.

The $MAPE$s for the ABML is statistically significantly lower ($p = 0.026$) than the standard NN at $\alpha = 0.5$. The mathematical model performs significantly better than the ABML.

One specific "cherry picked" example of the results is shown in Fig. 1. This example is chosen to exaggerate the performance improvement of the ABML compared to the NN. Not all results were this good, as summarised in Tab. 1.

4 Discussion

We have proposed a concept of using an agent based model to simulate data that is then used to train a meta learner. We have tested the performance of this technique on a simple sample problem involving the prediction of the future sales of a company based on a limited amount of data (10 training data points). We have compared the performance of the agent based meta learner (ABML) to both a standard neural network (NN), and a mathematical model (MM). The performance of the ABML is statistically significantly better than the NN at a significance threshold of $\alpha = 0.05$ as expected. However, unexpectedly, the performance of the ABML is significantly worse than the MM. This lower performance on the MM indicates significant room to improve the ABML technique. Some obvious improvements to the ABML would be training it for longer, and fine tuning the hyperparameters as the current iteration had somewhat arbitrary hyperparameters. While improvements to the ABML could be made, it is unlikely that it would outperform a MM on such a simple task. Despite this under performance, this paper has successfully proven the viability of an ABML and laid the groundwork for future work in refining the technique and applying it.

4.1 Strengths

There are a number of advantages to the ABML technique when compared to existing techniques. We outline a few of these.

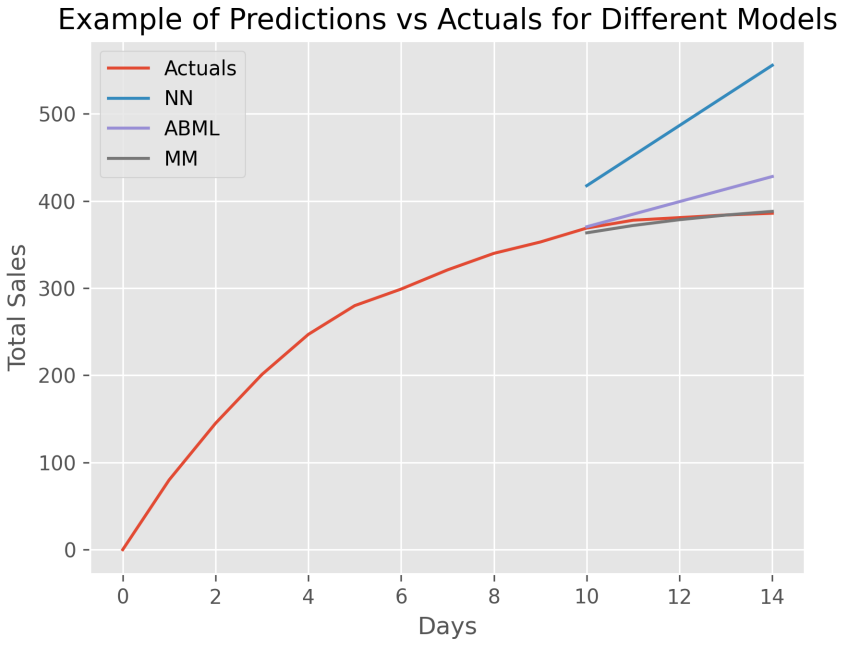


Figure 1: Example results. Picked to exaggerate the performance of the ABML compare to the NN.

ABMLs can outperform NNs on small datasets as demonstrated in this paper. With some further refinements the ABML technique may be a viable technique for applying deep learning techniques to small data sets.

ABMLs are simple to define compared to MMs. Building an ABML involves two steps, developing an ABM and defining a neural network. Defining a neural network is a well understood, and relatively simple task with modern software like PyTorch and TensorFlow. Defining an ABM in many circumstances is far simpler than defining a MM. ABMs can be built on intuition or on small scale scientific results, whereas mathematical models often require an expert in the field to corroborate with an applied mathematician or statistician. Additionally, justifying and explaining an ABM to s non-mathematician can be a simpler task than explaining a mathematical model, allowing for rapid feedback and a faster iteration cycle. Agent based models can also capture complex inter-agent interactions that can be complex or impossible to capture in a MM. This makes agent based models the only option for certain tasks.

4.2 ABML Limitations

While there are a number of advantages to ABMLs, there are some limitations, as outlined below.

1. ABMs can become computationally intensive when the number of agents and inter-agent interactions increases. Additionally, training a meta model is typically slower than training a single NN. This means that a NN will often be a better option if there is enough data to train it.

2. ABMs typically have to be analysed via simulations, and do not have the range of tools available to them as some mathematical models do. They also perform worse than MMs making an MM a better choice if one is available.

4.3 Paper Limitations

There are also some significant limitations to this paper that need to be explored/overcome in future work.

1. This paper only tested one simple model, further work needs to be done investigating other complex models and testing performance on them. It is the authors belief that ABMLs performance improvement over standard NNs will be even higher for complex systems.

2. The ABML and the NN were relatively "unrefined". As such, refinements need to be made to both the NN and the ABML to test their optimal performance. Some possible refinements include hyperparameters tuning, exploring alternate architectures, exploring other meta techniques than MAML, and feature engineering for the inputs.

3. This paper tested the performance of the ABML on data simulated from the exact same system as the training data (i.e. the same ABM), in the real world the system that generates the training data (i.e. the ABM) won't be the same as the system that

generates the testing data (i.e. the real world), as such testing performance on real world data sets is essential.

4.4 Future Work

Aside from addressing the above limitations, there are some interesting directions for further work.

One direction for future work is in the are of causal machine learning. It is often the goal to understand how changing x will impact y. Standard ML can often be limited in that it only understands the correlation between x and y. However, ABMs can be easily built to understand the causal effect of x on y, if this is done, x can be change in the ABM and the causal response on y can be observed. If the ABML is trained on data that involves the change in x that the researcher is interested in, then it is possible that the ABML will learn the causal connection between x and y. By a similar argument, an ABML may posses the ability to extrapolate to unseen values of x more successfully than a standard NN by simply simulating these unseen values of x in the ABM that is used to train the ABML.

Another direction for future work is in estimating uncertainty. The task is to make a specific prediction from the ABML and estimate how certain this prediction is. The confidence could be estimated by assessing the prediction accuracy of the ABML on a number of simulated data sets from the ABM. This prediction accuracy could then be converted to an uncertainty and translated back to the original prediction, acting as an uncertainty estimate. This would estimate the uncertainty inherent in the NN and the limited data it is trained on. It would not capture the uncertainty due to model misspecification. This uncertainty estimate would be particularly useful for extrapolation where statistical uncertainty estimates are often limited.

5 Conclusion

This paper aimed to assess the validity of using agent based models (ABMs) to generate training data for meta learners. ABMs offer a simpler alternative to mathematical models like differential equations, while meta-learning techniques provide improved data efficiency compared to standard neural networks and other deep learning methods. Our findings demonstrate that agent based meta learners (ABML) perform significantly better than neural networks, though their performance does not yet meet that of mathematical models. These results lays the groundwork for further research to refine the ABML technique, enabling its application to real world scenarios such as epidemiology, finance, businesses analytics, economics, policy making, and other social sciences. Overall, this work highlights the potential of ABMLs as a versatile modeling framework for limited data environments.

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